

SKY130A

PROJECT CARAC SKY130A

SKY130A Analog MOS Characterization

Device Characterization and gm/ID Design Data for Analog IC Design

PDK	SKY130A
Tools	Xschem · ngspice · IIC-OSIC-TOOLS
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1 Project Overview

1.1 Objective

The purpose of this project is to characterize a selected set of SKY130A MOS devices and generate practical device data for subsequent analog IC design.

The objective is not to reproduce MOS transistor theory or to exhaustively characterize every device available in the PDK. Instead, the characterization focuses on quantities that are directly useful when selecting a device, estimating its operating point, and performing transistor sizing.

The characterized device families are:

- the standard 1.8 V NMOS and PMOS devices;
- the `nfet_03v3_nvt` native NMOS;
- the `nfet_05v0_nvt` native NMOS.

Additional device families can be characterized later when required by a specific circuit architecture.

1.2 Characterization Scope

The characterization concentrates on the quantities most useful during analog design:

- drain-current transfer characteristics;
- output-characteristic families;
- threshold voltage V_{TH} ;
- saturation voltage V_{DSAT} ;
- transconductance efficiency g_m/I_D ;
- effective MOS gate capacitance.

The main purpose of the generated curves is to provide reusable sizing and biasing references rather than to illustrate fundamental MOS operation.

1.3 Characterization Flow

The characterization flow uses Xschem testbenches and the SKY130A device models simulated with ngspice inside the IIC-OSIC-TOOLS environment.

DC and AC sweeps are exported as numerical data and post-processed with Python to generate the figures presented in this report.

Unless otherwise stated, the results correspond to the nominal TT process corner at 27 °C.

1.4 Device Quick Reference

Table 1 provides a compact reference for the characterized devices. It is intended to be the first source of device-level information during future circuit design.

Table 1 Quick reference for the characterized SKY130A MOS devices at TT and 27 °C.

Device	Ref. W/L ($\mu\text{m}/\mu\text{m}$)	$ V_{GS} _{\text{max}}$ (V)	$ V_{DS} _{\text{max}}$ (V)	V_{TH} (V)	$ V_{DSAT} $ (V)	$ I_D /W$ (mA/ μm)
nfet_01v8	10/0.5	1.95	1.95	+0.611	0.573	0.220
pfet_01v8	10/0.5	1.95	1.95	-0.992	0.695	0.0419
nfet_03v3_nvt	10/0.5	3.3	3.3	-0.0133	1.032	0.512
nfet_05v0_nvt	10/2	5.5	5.5	+0.0529	3.159	0.341

Design Note

The voltage limits reported in Table 1 correspond to the documented SPICE-model operating ranges and should not be interpreted as device absolute-maximum ratings. The documented ranges are taken from the SKY130A device documentation [1].

The extracted quantities use the reference transfer-characterization biases: $|V_{GS}| = 1.8\text{ V}$ and $|V_{DS}| = 0.9\text{ V}$ for the standard 1.8 V devices, $V_{GS} = 3.3\text{ V}$ and $V_{DS} = 1.65\text{ V}$ for the nfet_03v3_nvt, and $V_{GS} = 5\text{ V}$ and $V_{DS} = 2.5\text{ V}$ for the nfet_05v0_nvt.

2 Characterization Methodology

2.1 Simulation Environment

The characterization flow uses:

- Xschem for schematic capture and simulation testbenches;
- ngspice for DC and AC analyses;
- SKY130A compact device models;
- Python for numerical post-processing and plotting.

Unless otherwise stated, simulations use the nominal TT model at 27 °C.

2.2 DC Characterization

Two complementary analyses are used for the standard MOS devices:

- a gate-voltage sweep to obtain transfer and operating-point data;
- a drain-voltage sweep to obtain output-characteristic families.

The corresponding NMOS and PMOS transfer testbenches are shown in Figure 1.

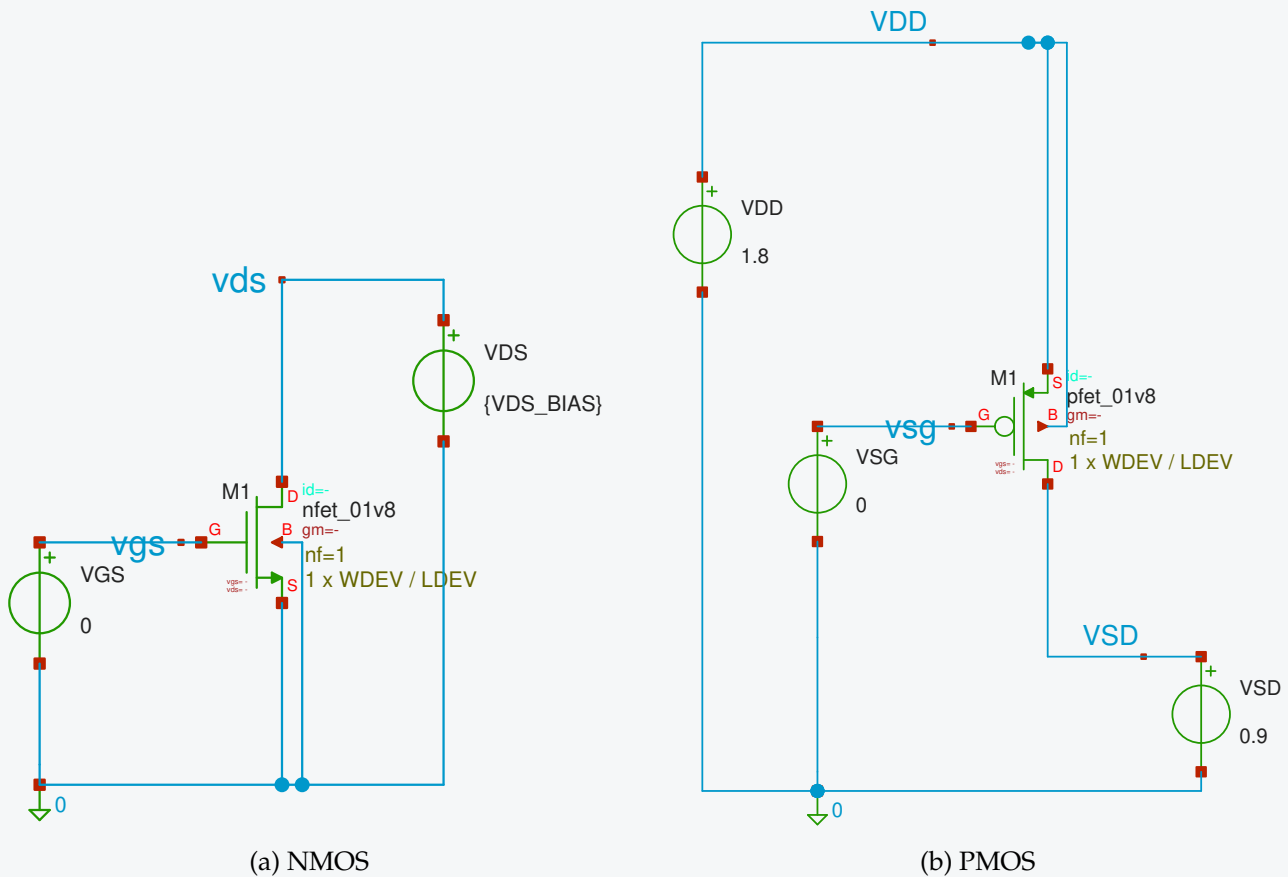


Figure 1 Transfer-characterization testbenches for the standard 1.8 V MOS devices.

The output-characteristic benches are shown in Figure 2.

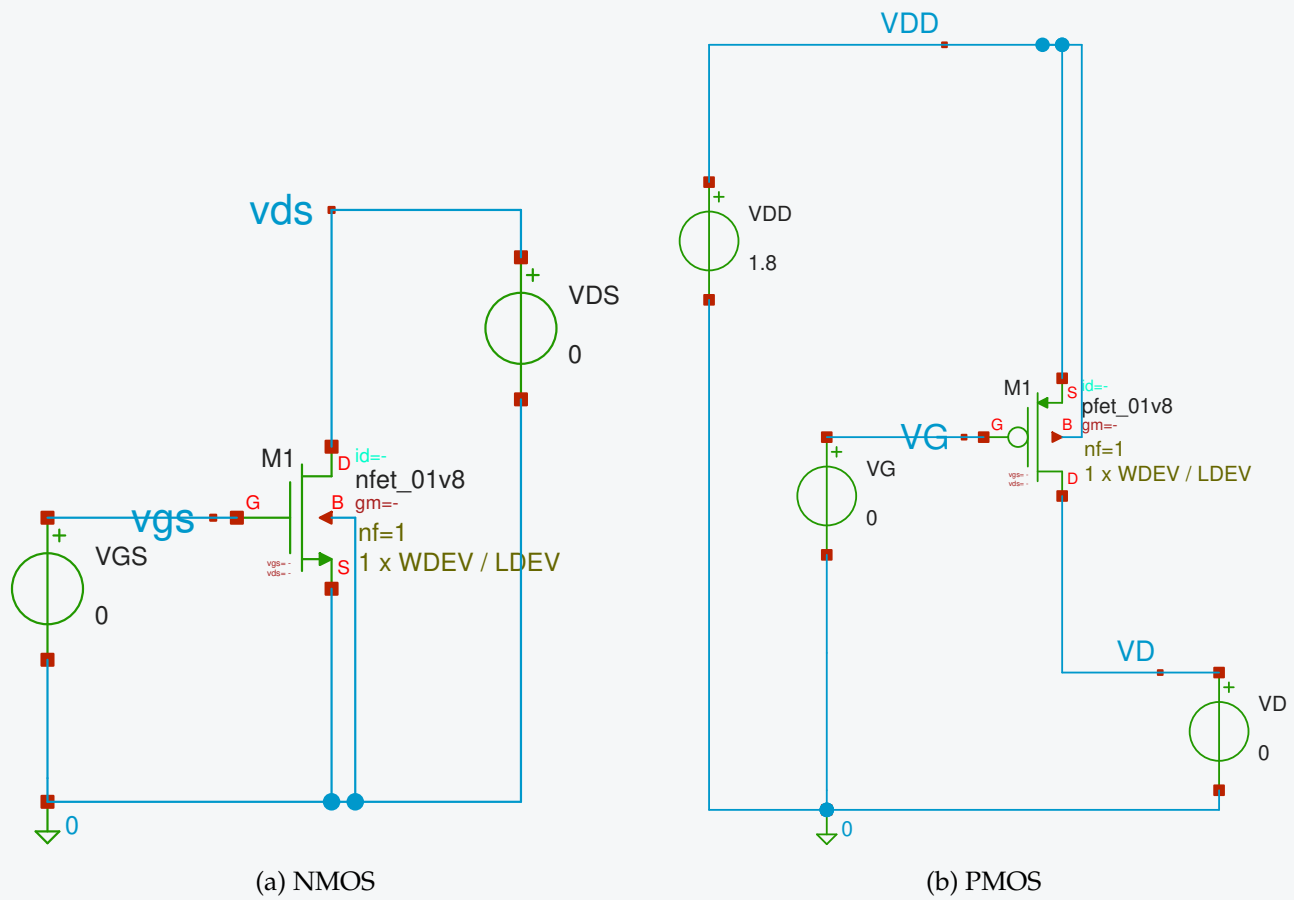


Figure 2 Output-characteristic testbenches for the standard 1.8 V MOS devices.

2.3 Sweep Conditions

The principal characterization conditions are summarized in Table 2.

Table 2 Principal standard-device characterization sweeps.

Analysis	Swept quantity	Fixed / stepped condition
NMOS transfer	$V_{GS} : 0 \rightarrow 1.8 \text{ V}$	$V_{DS} = 0.9 \text{ V}$
PMOS transfer	$V_{SG} : 0 \rightarrow 1.8 \text{ V}$	$V_{SD} = 0.9 \text{ V}$
NMOS output	$V_{DS} : 0 \rightarrow 1.8 \text{ V}$	$V_{GS} = 0.6, 0.8, \dots, 1.8 \text{ V}$
PMOS output	$V_{SD} : 0 \rightarrow 1.8 \text{ V}$	$V_{SG} = 0.6, 0.8, \dots, 1.8 \text{ V}$

2.4 Extracted Design Quantities

The characterization is organized around quantities that can be reused directly during transistor sizing.

Table 3 Purpose of the principal extracted quantities.

Quantity	Design use
$I_D(V_{GS})$	Current capability and gate-bias selection.
$I_D(V_{DS})$	Output behaviour, saturation region and available current.
V_{TH}	Threshold reference and geometry dependence.
V_{DSAT}	Drain-source headroom required for saturation.
g_m/I_D	Direct transconductance-efficiency reference for analog sizing.
C_G	Estimate of the capacitive cost associated with MOS gates.

For the drain-current extraction, the ngspice voltage-source current convention is accounted for using

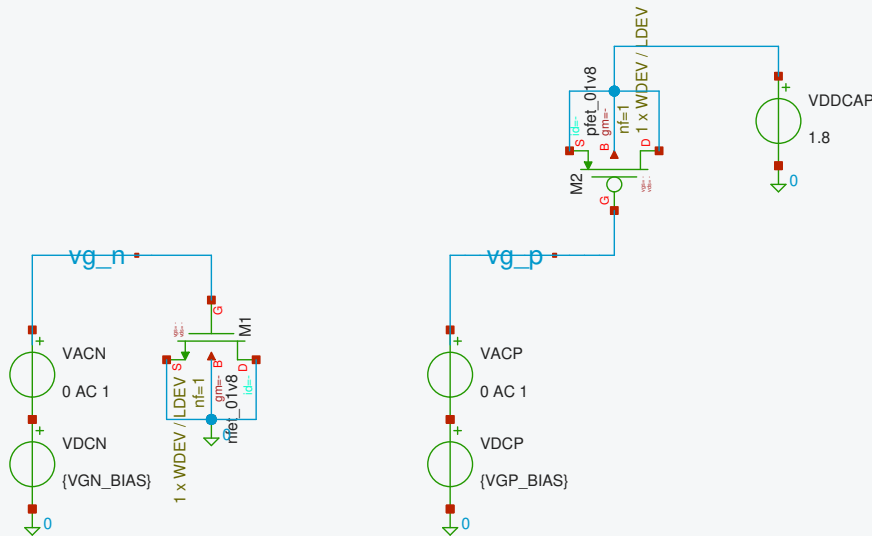
$$I_D = -I(V_{DS}). \quad (1)$$

Where available, V_{TH} and V_{DSAT} are taken directly from the compact-model operating-point information.

2.5 MOS Capacitance Characterization

The standard NMOS and PMOS devices are also evaluated as two-terminal MOS capacitors.

For the NMOS, drain, source and bulk are shorted to ground. For the PMOS, the same terminals are connected to the 1.8 V supply.

**Figure 3** Combined NMOS and PMOS gate-capacitance characterization testbench.

A unit-magnitude small-signal AC excitation is applied to the gate at 1 MHz. The equivalent gate capacitance is obtained from

$$C_G = \frac{|\text{Im}(I_G)|}{2\pi f}. \quad (2)$$

The retained results are C_G versus capacitor voltage, used to quantify bias dependence, and $C_G/(WL)$ versus channel length, used to quantify geometry dependence and estimate effective capacitance density.

3 Standard 1.8-V MOS Characterization

3.1 Device Under Test

The standard 1.8 V devices are

`sky130_fd_pr__nfet_01v8` and `sky130_fd_pr__pfet_01v8`.

Unless otherwise stated,

$$W = 10\,\mu\text{m}, \quad L = 0.5\,\mu\text{m}. \quad (3)$$

All results correspond to TT at 27 °C.

3.2 Drain Current Characteristics

3.2.1 Transfer Characteristics

The transfer characteristic provides a direct reference for gate bias and available current at

$$|V_{DS}| = 0.9\,\text{V}. \quad (4)$$

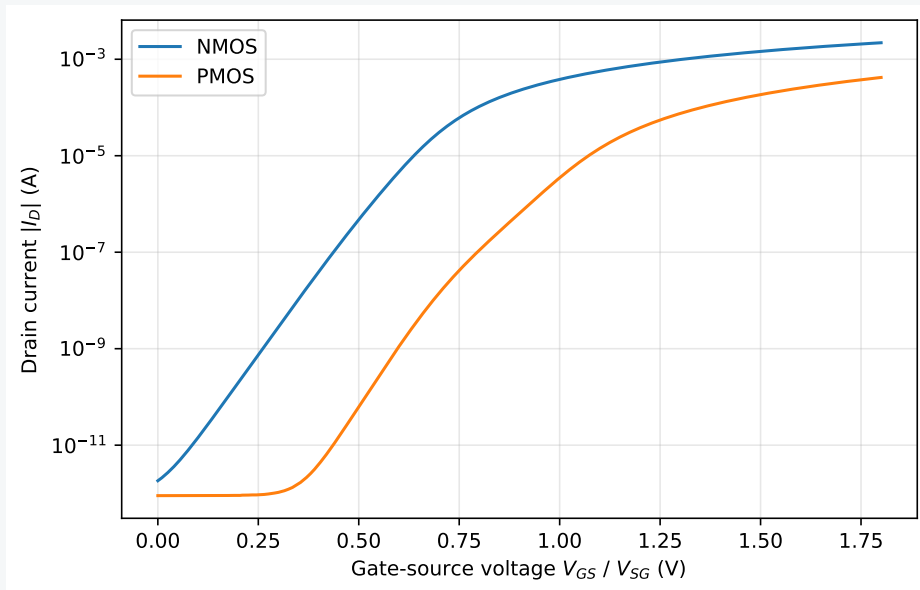


Figure 4 Drain-current transfer characteristics of the standard 1.8 V NMOS and PMOS devices.

For equal geometry and gate-voltage magnitude, the NMOS provides significantly higher current than the PMOS.

3.2.2 Output Characteristics

The output-characteristic families provide a direct reference for current capability and saturation behaviour.

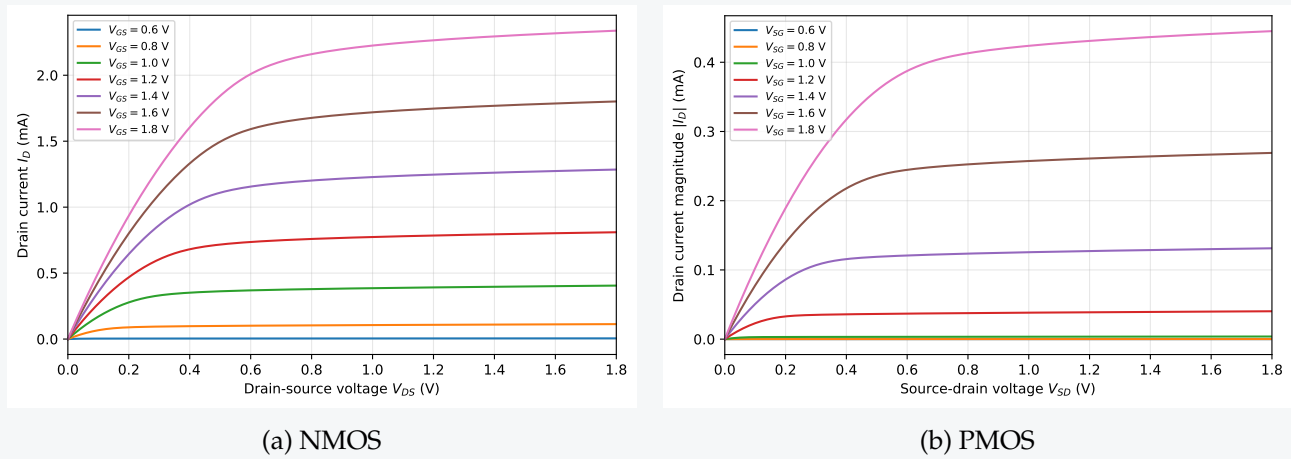


Figure 5 Output-characteristic families of the standard 1.8 V MOS devices for gate-voltage magnitudes from 0.6 V to 1.8 V.

At the highest gate bias, the NMOS current is approximately five times larger than the PMOS current for the same geometry.

3.3 Threshold and Saturation Voltages

3.3.1 Threshold Voltage

At $|V_{DS}| = 0.9$ V and $V_{SB} = 0$, the compact-model operating-point data give approximately:

Table 4 Threshold voltages for the reference 1.8 V devices.

Device	W/L ($\mu\text{m}/\mu\text{m}$)	$ V_{TH} $
NMOS	10/0.5	0.61 V
PMOS	10/0.5	0.99 V

The geometry dependence of V_{TH} and the bias dependence of V_{DSAT} are shown together in Figure 6.

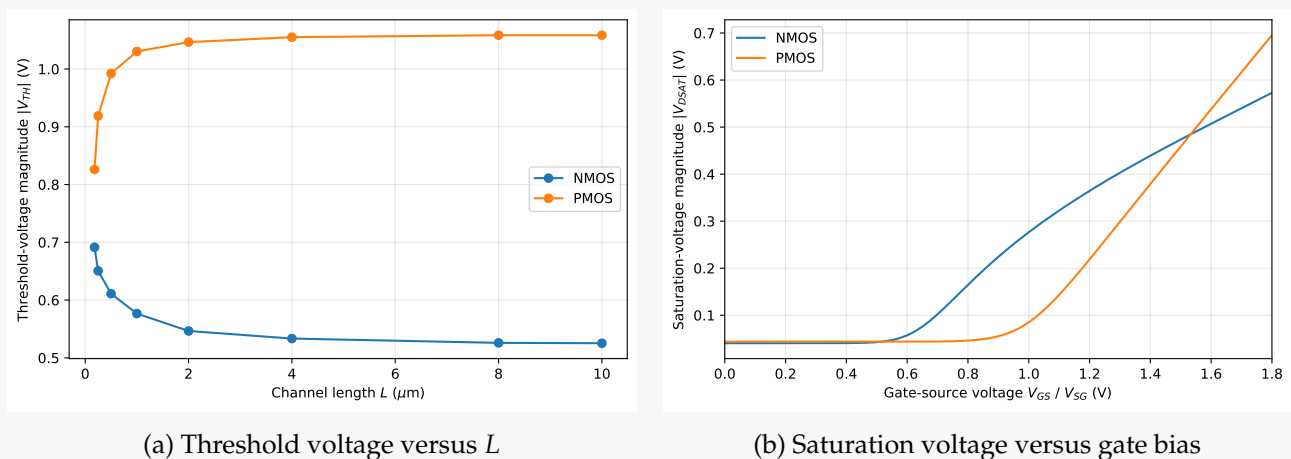


Figure 6 Threshold- and saturation-voltage characteristics of the standard 1.8 V devices.

The NMOS threshold decreases with increasing channel length, whereas the PMOS threshold magnitude increases toward approximately 1.06 V.

The V_{DSAT} characteristic provides a direct headroom reference for future bias selection.

3.4 Transconductance Efficiency

The g_m/I_D characteristic is retained as the main sizing-oriented result. Drain current is normalized by width so that the curve can be used independently of the reference transistor width.

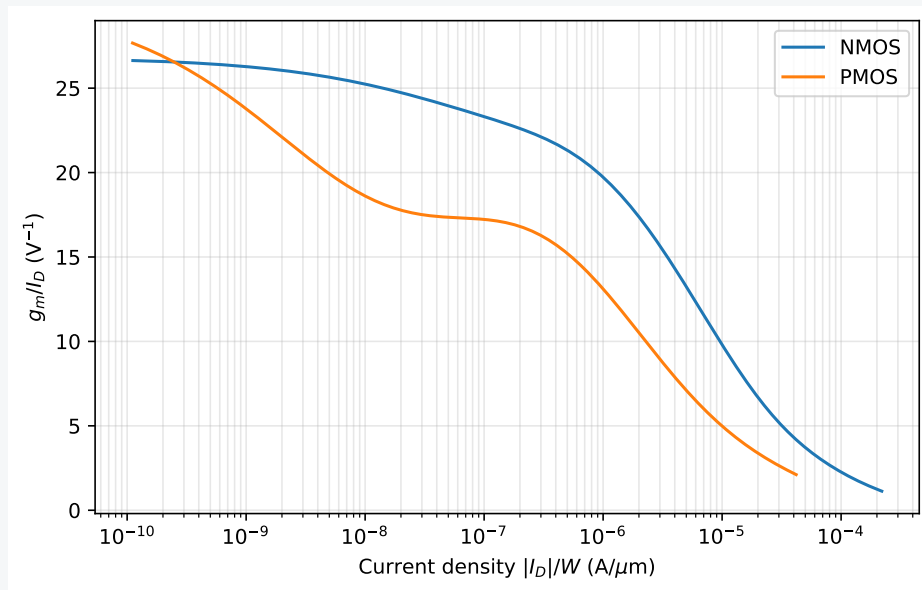


Figure 7 Transconductance efficiency g_m/I_D versus drain-current density $|I_D|/W$.

3.5 Gate Capacitance

The effective gate capacitance is characterized at 1 MHz as a function of bias and geometry.

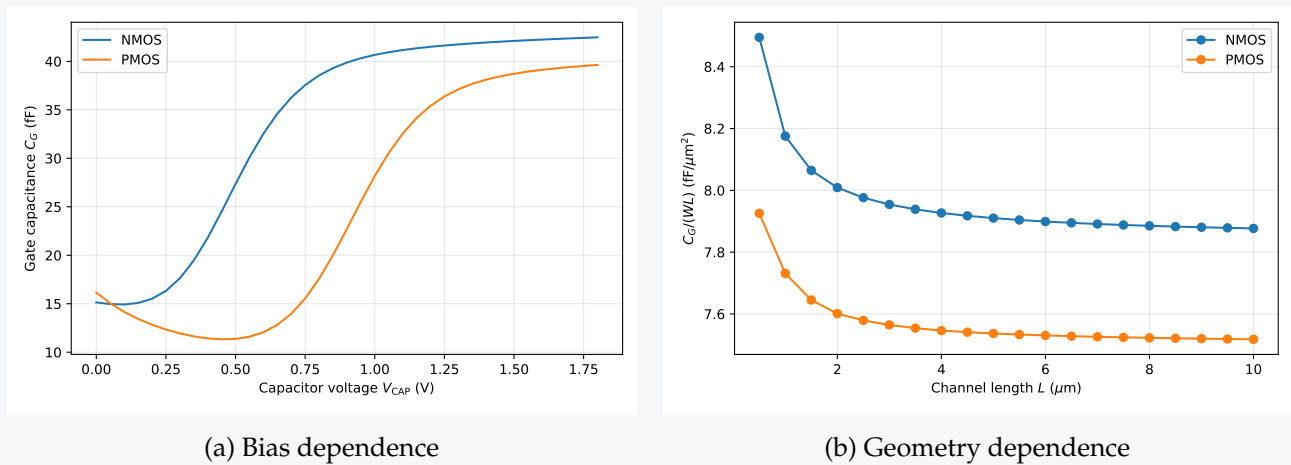


Figure 8 Effective gate-capacitance characteristics of the standard 1.8 V NMOS and PMOS devices.

At long channel lengths, the extracted capacitance density approaches approximately $7.9 \text{ fF } \mu\text{m}^{-2}$ for the NMOS and $7.5 \text{ fF } \mu\text{m}^{-2}$ for the PMOS.

The larger normalized capacitance observed for short devices reflects the greater relative contribution of edge and overlap capacitances.

4 Native NMOS Characterization

4.1 Devices Under Test

Two native NMOS devices are characterized:

`sky130_fd_pr__nfet_03v3_nvt` and `sky130_fd_pr__nfet_05v0_nvt`.

Their reference conditions are summarized in Table 5.

Table 5 Native-device reference geometries and transfer-bench drain biases.

Device	Reference W/L	Transfer V_{DS}
<code>nfet_03v3_nvt</code>	10/0.5 $\mu\text{m}/\mu\text{m}$	1.65 V
<code>nfet_05v0_nvt</code>	10/2 $\mu\text{m}/\mu\text{m}$	2.5 V

The larger channel length used for the `nfet_05v0_nvt` is imposed by the restricted geometry bins available in its SKY130A compact model.

4.2 3.3-V Native NMOS

Because the native threshold is close to zero, the transfer sweep is extended below ground to capture the turn-off region.

The characterization benches are shown in Figure 9.

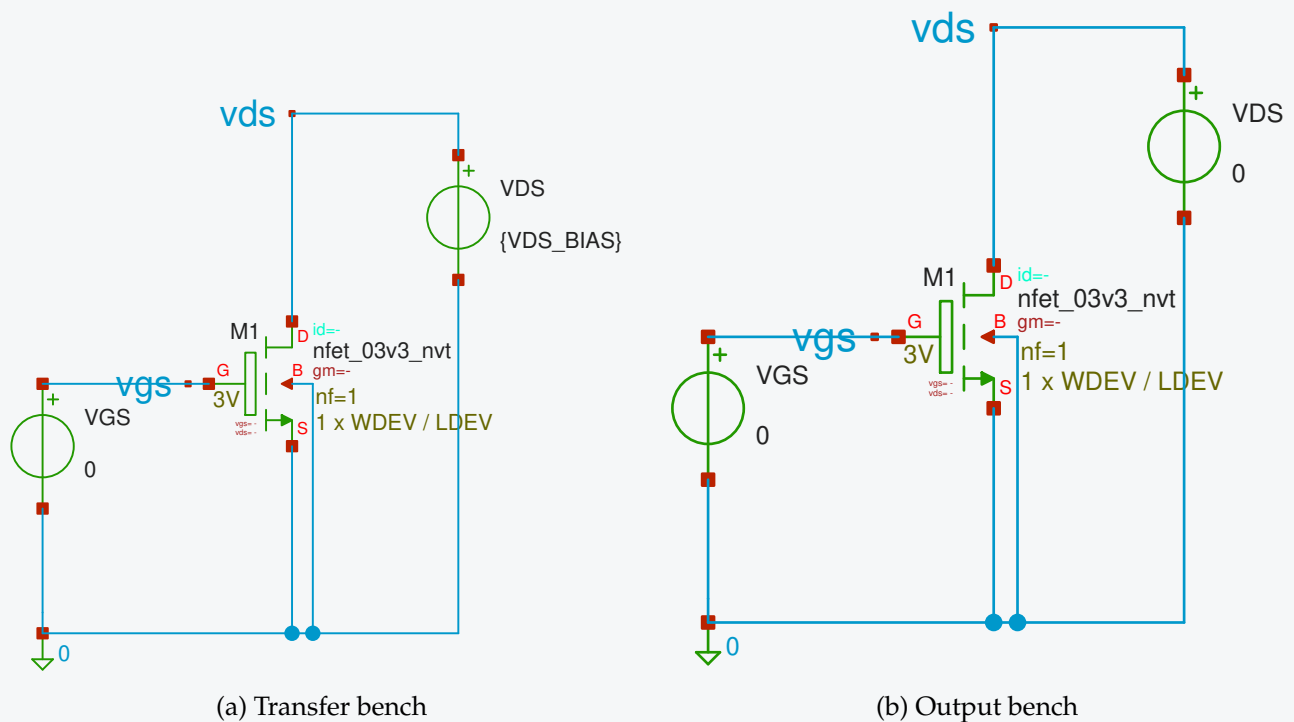


Figure 9 Characterization testbenches for the `nfet_03v3_nvt`.

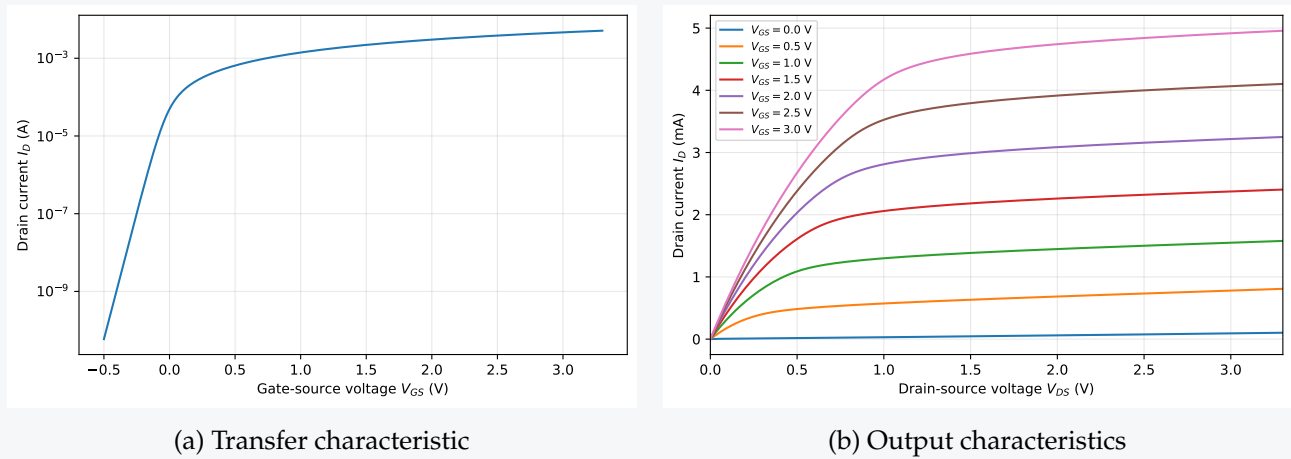


Figure 10 Drain-current characteristics of the `nfet_03v3_nvt`.

The device exhibits the expected native behaviour, with an extracted threshold voltage of approximately $V_{TH} = -13.3$ mV for the reference geometry. Consequently, measurable drain current is already present at $V_{GS} = 0$.

4.3 5-V Native NMOS

The `nfet_05v0_nvt` is characterized using

$$W = 10\ \mu\text{m}, \quad L = 2\ \mu\text{m}. \quad (5)$$

This geometry is explicitly supported by the available compact-model bins.

For the reference geometry, the extracted threshold voltage is approximately $V_{TH} = 52.9$ mV, confirming the near-zero-threshold behaviour of this device.

The characterization benches are shown in Figure 11.

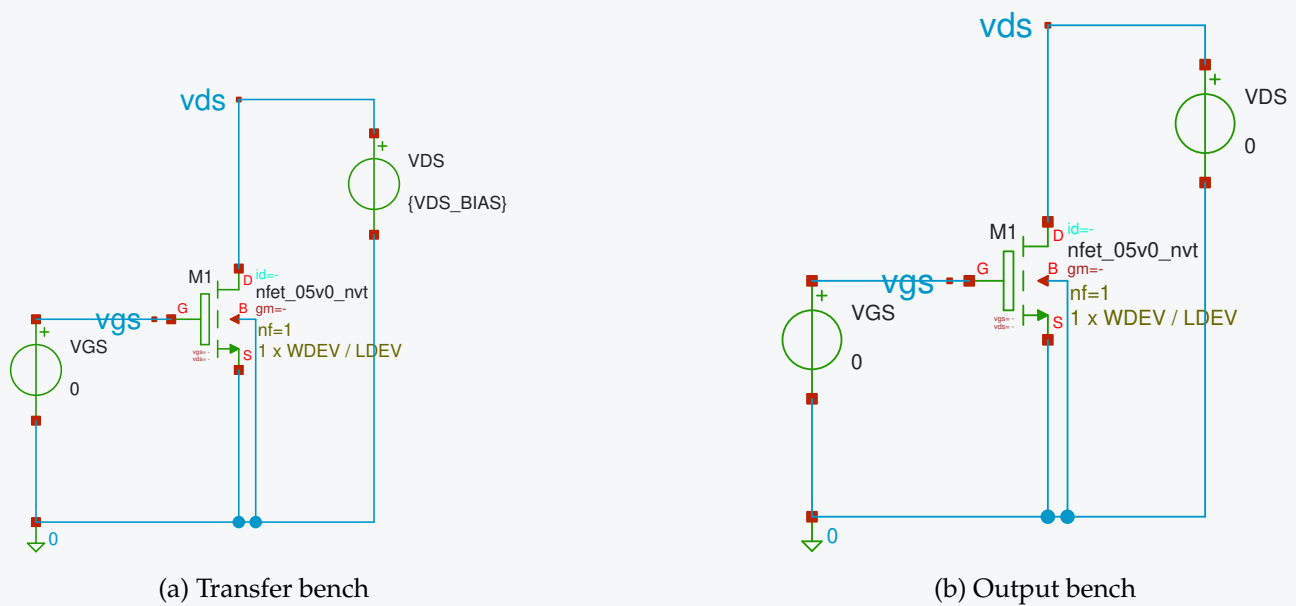


Figure 11 Characterization testbenches for the `nfet_05v0_nvt`.

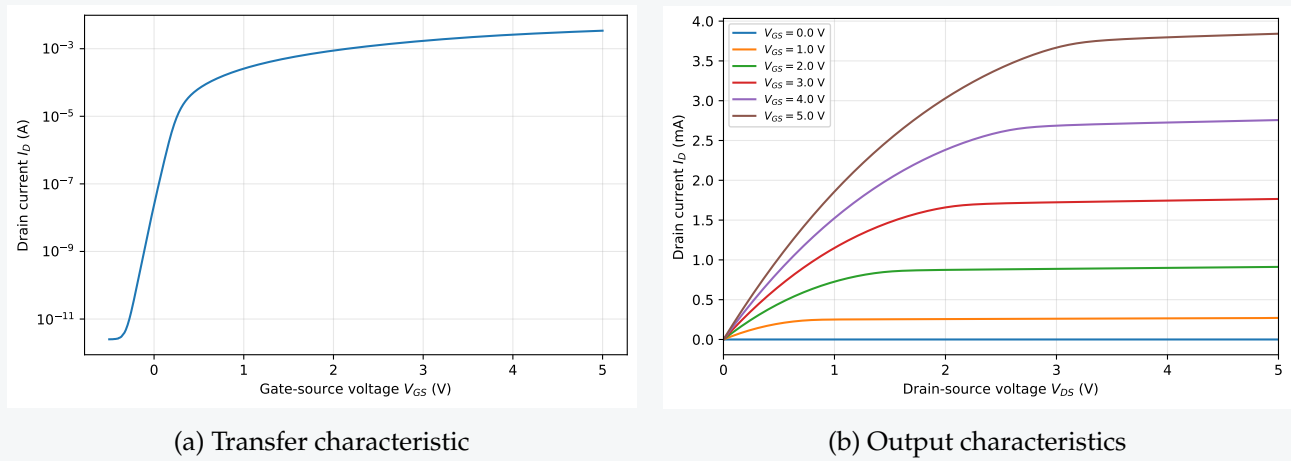


Figure 12 Drain-current characteristics of the `nfet_05v0_nvt`.

4.4 Transconductance Efficiency

The g_m/I_D characteristics are retained as sizing references for the two native devices.

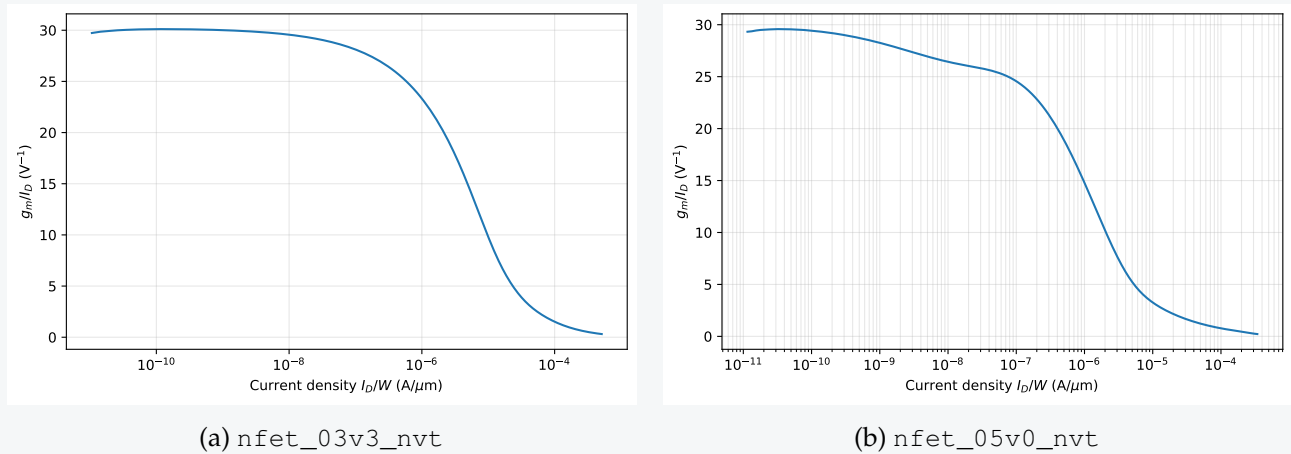


Figure 13 Transconductance efficiency versus drain-current density for the characterized native NMOS devices.

4.5 Native Device Comparison

Both native devices provide low-threshold operation for circuits where the standard 1.8 V NMOS threshold would impose excessive gate-voltage headroom.

The `nfet_03v3_nvt` provides the lower-voltage native option, while the `nfet_05v0_nvt` extends the usable voltage range but is available only for a restricted set of compact-model geometries.

The transfer, output, and g_m/I_D characteristics therefore provide the main information required when selecting between the two devices.

5 Conclusion

This project establishes a compact characterization reference for the SKY130A MOS devices most relevant to the planned analog design work.

The standard 1.8 V NMOS and PMOS were characterized in terms of drain current, threshold and saturation voltages, transconductance efficiency, and gate capacitance.

The `nfet_03v3_nvt` and `nfet_05v0_nvt` were additionally characterized to provide low-threshold alternatives for circuits requiring native-device operation.

The report intentionally concentrates on practical design quantities rather than providing an exhaustive characterization of the complete PDK. Device families such as the high-voltage MOS devices can be added when a future circuit—for example a CIS architecture—actually requires them.

The quick-reference table in Section 1.4 is intended to provide the main entry point for future device selection, while the detailed plots remain available when bias- or geometry-dependent information is required.

References

- [1] SkyWater Technology Foundry. *SKY130 PDK Device Details*. URL: <https://skywater-pdk.readthedocs.io/en/main/rules/device-details.html> (visited on 09/16/2026).